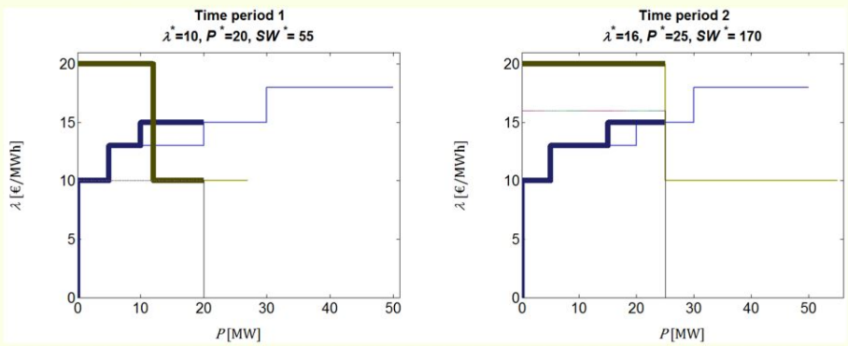
Exercise 6: (MPA) model.

Let's consider the instance of the Multiperiod Auction problem (MPA) with two periods ($\mathcal{T} = \{1,2\}$), tow generation units and a single demand units associated to the following data:

Generation units	$P_{G_i}^{max}$ $[MW]$	$P_{G_i}^{min}$ $[MW]$	$R_{G_i}^{up}$ $[MW]$	$R_{G_i}^{down}$ $[MW]$	$P_{G_i}^{0}$ $[MW]$	$P_{G_{ikt}}^{b}[MW]\forall t$		$\lambda_{G_{ikt}}^{b}\left[\frac{€}{MWh}\right]$	
						$k=1$	$k=2$	$k=1$	$k=2$
1	20	5	5	5	10	5	15	10	13
2	30	10	10	10	15	10	20	15	18

Demand bid function	$P_{D_{ikt}}^{b}[MW]$		$\lambda_{D_{ikt}}^{b}[€/MWh]$	
	$k=1$	$k=2$	$k=1$	$k=2$
$t=1$	12	15	20	10
$t=2$	25	30	20	10

a) Find the optimal market equilibrium with GAMS and check that it coincides with the following results:

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---- VAR PG  Matched generation of each block [MW]

            LOWER     LEVEL     UPPER    MARGINAL

t1.G1.b1      .       5.000     +INF        .
t1.G1.b2      .       5.000     +INF        .
t1.G2.b1      .      10.000     +INF        .
t1.G2.b2      .        .        +INF        .
t2.G1.b1      .       5.000     +INF        .
t2.G1.b2      .      10.000     +INF        .
t2.G2.b1      .      10.000     +INF        .
t2.G2.b2      .        .        +INF        .
```

```
---- VAR PD  Matched demand of each block [MW]

            LOWER     LEVEL     UPPER    MARGINAL

t1.D1.b1      .      12.000     +INF        .
t1.D1.b2      .       8.000     +INF        .
t2.D1.b1      .      25.000     +INF        .
t2.D1.b2      .        .        +INF      -6.0
```

```
SETS
g Generation units / G1*G2/
bg Blocks           /b1*b2/;

PARAMETERS
pg_max(g) Max. active power generation [MW] /G1 20, G2 30/
pg_min(g) Min. active power generation [MW] /G1  5, G2 10/
ru(g)     Ramp-up limit [MW:h]              /G1  5, G2 10/
rd(g)     Ramp-down limit [MW]              /G1  5, G2 10/
p0(g)     Initial power output [MW]         /G1 10, G2 15/;

TABLE pbg(t,g,bg) Power [MW] of the generation bid function
       b1  b2
t1.G1   5  15
t1.G2   10 20
t2.G1   5  15
t2.G2   10 20;

TABLE lbg(t,g,bg) Prices [eur:MWh] of the generation bid function
      b1   b2
t1.G1  10 13
t1.G2  15 18
t2.G1  10 13
t2.G2  15 18;

* Demand units:

SETS
d Generation units / D1 /
bd Blocks          / b1*b2 /;

TABLE pbd(t,d,bd) Power [MW] of the demand bid function
      b1 b2
t1.D1  12 15
t2.D1  25 30;

TABLE lbd(t,d,bd) Prices [�:MWh] of the generation bid function
      b1 b2
t1.D1 20 10
t2.D1 20 10 ;
```

```
                        LOWER          LEVEL          UPPER         MARGINAL

---- VAR SW             -INF          225.0000        +INF             .

  SW  Social welfare (objective function)

---- VAR PG  Matched generation of each block [MW]

                  LOWER          LEVEL          UPPER         MARGINAL

t1.G1.b1            .            5.0000         +INF             .
t1.G1.b2            .            5.0000         +INF             .
t1.G2.b1            .           10.0000         +INF             .
t1.G2.b2            .              .            +INF             .
t2.G1.b1            .            5.0000         +INF             .
t2.G1.b2            .           10.0000         +INF             .
t2.G2.b1            .           10.0000         +INF             .
t2.G2.b2            .              .            +INF             .

---- VAR PD  Matched demand of each block [MW]

                  LOWER          LEVEL          UPPER         MARGINAL

t1.D1.b1            .           12.0000         +INF             .
t1.D1.b2            .            8.0000         +INF             .
t2.D1.b1            .           25.0000         +INF             .
t2.D1.b2            .              .            +INF          -6.0000
```

b) Explain the reason why the second block of the first generation unit at $t = 2$ cannot be completely matched.

The rampup constraint of generator 1 is limited to 5MW, in t=2 the generator could only rampup from 10 to 15MW eventough more power was offered in the block.

```
---- VAR PG  Matched generation of each block [MW]

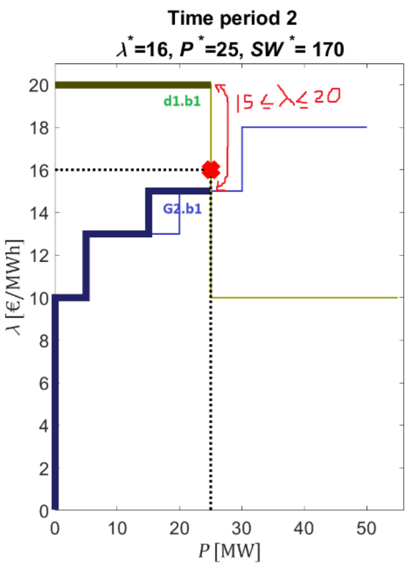
                  LOWER          LEVEL          UPPER

t1.G1.b1            .            5.0000         +INF
t1.G1.b2            .            5.0000         +INF
t1.G2.b1            .           10.0000         +INF
t1.G2.b2            .              .            +INF
t2.G1.b1            .            5.0000         +INF
t2.G1.b2            .           10.0000         +INF
t2.G2.b1            .           10.0000         +INF
t2.G2.b2            .              .            +INF
```

10

15

c) The price equilibrium conditions state that in an optimal market clearing the equilibrium price is set by the bid price of the marginal unit, that is, the price of the last partially matched production or demand block. Check if this principle is satisfied by the optimal market clearing found by GAMS for the two auctions, paying special attention to the second auction.



As both generation and demand blocks are filled there is not a marginal unit , the market price could be any integer value between 15 (block price of demand) and 20 (block price of generation).

SW would be the exact same area for any market clearing within those values.

GAMS chose 16.

d) Explain the reasons why in the first auction generation unit 2 has a block matched below their bid price (underpaid).

The last block partially filled sets the market price, this time corresponds with a demand block instead of a generator block.

T1.D1.b2 sets the market price and not the fully filled block t1.g2.b1